

Structure-Aware Molecular Machine Learning for Acetylcholinesterase Inhibitory Potency Prediction

AChE QSAR Modelling with Molecular Fingerprints,
Physicochemical Descriptors, Scaffold-Aware Validation,
Ensemble Learning, and Graph-Based Representation

Scientific project report

Target: ChEMBL220 (acetylcholinesterase)

Endpoint: $IC_{50} \rightarrow pIC_{50}$ regression

Primary dataset after current curation: 4,571 molecular records

Repository:

<https://github.com/TheHashiramaSenju/AChE-QSAR-modelling>

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Computational Biology · Cheminformatics · Molecular Machine Learning

Independent computational study

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Contents

Document Status and Evidence Policy	5
1 Abstract	5
2 Introduction	6
2.1 Acetylcholinesterase as a Molecular Target	6
2.2 Structure–Activity Relationships	6
2.3 From an Educational Workflow to a Structured Investigation	7
2.4 Research Problem	7
2.5 Research Questions	7
2.6 Working hypotheses	8
3 Dataset and Bioactivity Data	8
3.1 Target Selection	8
3.2 Why the Dataset Is Not Equivalent to a Single Experiment	8
3.3 Raw Dataset Composition	8
3.4 Activity Endpoint and Target Transformation	9
4 Data Curation	9
4.1 Curation Philosophy	9
4.2 Curation Audit	10
4.3 Structure Standardization	10
4.4 Replicate Aggregation	10
4.5 Equality Relation	11
4.6 Activity Cutoff	11
4.7 Structure Sanity Warnings	11
5 Molecular Representation	12
5.1 Representation as a Scientific Choice	12
5.2 Morgan Fingerprints	12
5.3 RDKit Descriptor Representation	13
5.4 Combined Representation	14
6 Scaffold-Aware Partitioning	14
6.1 Motivation	14

6.2	Implemented Allocation	14
6.3	A Critical Property of the Current Split	15
6.4	Target Distribution Across the Saved Split	15
7	Classical Machine Learning	16
7.1	Experimental Matrix	16
7.2	Random Forest	16
7.3	XGBoost	16
7.4	LightGBM	16
7.5	Optuna Objective	17
8	Original Classical Results	17
8.1	Held-Out Performance	17
8.2	Representation Effects	18
8.3	Model Effects	18
9	Hyperparameter Optimization	19
9.1	What Was Actually Run	19
9.2	Saved Best Parameters	19
10	Graph Neural Network	20
10.1	Motivation	20
10.2	Atom Features	20
10.3	Edge Features	20
10.4	Global Descriptors	21
10.5	Message Passing	21
10.6	Graph Readout	21
10.7	Training	22
10.8	GNN Result	22
11	Clean GNN–Classical Comparison Audit	22
11.1	Purpose	22
11.2	Clean Comparison Results	23
11.3	Why the GNN Result Is Scientifically Useful	23
12	Results and Interpretation	24
12.1	What the Best Model Actually Demonstrates	24

12.2 Correlation Versus Error	24
12.3 Why RMSE Matters	24
13 Error Analysis	25
13.1 Residual Definition	25
13.2 Available Diagnostic Outputs	25
13.3 Activity Cliffs	25
13.4 Applicability Domain	26
14 Scientific Interpretation of Molecular Representations	26
14.1 Morgan Fingerprints: What They Give the Model	26
14.2 RDKit Descriptors: What They Add	26
14.3 Why the Combined Representation Helps in This Dataset	27
15 Reproducibility	27
15.1 Repository Organization	27
15.2 Recorded Software Environment	27
15.3 Seeds	28
16 Implementation Audit and Required Corrections	28
16.1 Issue 1: Feature Count in Original Metric CSVs	28
16.2 Issue 2: Scaffold Allocation Bias	28
16.3 Issue 3: Disconnected Structures	29
16.4 Issue 4: Morgan Fingerprint Function Signature	29
16.5 Issue 5: Activity Threshold Expression	29
16.6 Issue 6: GNN Edge-Feature Documentation	29
16.7 Issue 7: GNN Is Not Graph-Only	29
16.8 Issue 8: Single Fixed Split	29
17 Limitations	30
17.1 Dataset Limitations	30
17.2 Endpoint Limitations	30
17.3 Representation Limitations	30
17.4 Validation Limitations	30
17.5 Model Limitations	30
17.6 Biological Limitations	30

18 Conclusions	30
19 Future Work	31
19.1 Immediate Methodological Corrections	31
19.2 Chemical Error Analysis	32
19.3 Repeated Scaffold Validation	32
19.4 Graph Modelling	32
19.5 External Validation	33
19.6 Mechanistic Extensions	33
20 Scientific Decision Rationale	33
20.1 Why ChEMBL220 and IC ₅₀ ?	33
20.2 Why retain only equality relations?	33
20.3 Why median aggregation and MAD = 1.0?	33
20.4 Why 10,000 nM?	34
20.5 Why Morgan radius 2 and 1,024 bits?	34
20.6 Why these 14 descriptors?	34
20.7 Why scaffold separation?	34
20.8 Why five-fold GroupKFold for the original classical search?	34
20.9 Why Optuna and TPE?	34
20.10 Why the GNN architecture?	35
20.11 Why CPU and seed 50?	35
20.12 What is deliberately not claimed?	35
A Supplementary Table: Original Metrics	35
B Supplementary Table: Clean GNN/Classical Audit	36
C Diagnostic Figures	36
D Reproducibility Record	38
D.1 Direct Code Traceability	38
E Complete Model Diagnostics	39
F Final Audit Record	42

Document Status and Evidence Policy

This report is intentionally written as an evidence-grounded scientific manuscript rather than as a generic description of QSAR modelling. The repository, source code, saved audit outputs, model metric files, prediction files, and training logs are treated as the primary evidence base.

Three classes of statements are kept separate:

1. **Implemented facts:** directly supported by the source code.
2. **Measured results:** directly supported by saved CSV, JSON, or audit outputs.
3. **Scientific interpretation:** interpretation of those measurements that remains subject to the limitations of the experimental design.

Scientific rationale is stated explicitly throughout the report. Where a rationale is an engineering judgement rather than an independently isolated experimental optimum, it is identified as such rather than presented as a measured fact. All numerical results are tied to saved project outputs, and code-reference markers link major implementation decisions directly to the corresponding GitHub source lines.

Important methodological audit finding

The repository contains two relevant generations of classical-model results. The original model logs report test sets of 915 molecules and use a 5-fold scaffold-grouped CV objective. A later dedicated GNN/classical comparison audit uses the same 915-molecule test partition but reserves an additional 548-molecule validation set from the original training portion, leaving 3,108 molecules for model training. The latter audit also corrects feature-count accounting and performs training-fold imputation for missing descriptors. These result sets are therefore **not numerically interchangeable** and are reported separately in this manuscript.

1 Abstract

This study develops and audits a structure-aware machine-learning pipeline for predicting acetylcholinesterase (AChE) inhibitory potency from molecular structure. Bioactivity records associated with the human AChE target ChEMBL220 were retrieved from ChEMBL and subjected to a staged curation workflow involving validity filtering, salt removal, molecular identity assignment through InChIKey generation, removal of records flagged as potential duplicates, activity-unit normalization, pIC₅₀ transformation, replicate aggregation, exact duplicate removal, equality-relation filtering, and an activity cutoff corresponding to IC₅₀ ≤ 10 000 nM. The resulting current cohort contains 4,571 unique molecular identities with finite pIC₅₀ values.

Two engineered molecular representations were investigated: 1024-bit Morgan circular fingerprints with radius 2 and a 14-dimensional RDKit descriptor vector. These were evaluated individually and in combination using Random Forest, XGBoost, and LightGBM regressors. Hyperparameters were optimized with Optuna under scaffold-grouped cross-validation, while the final held-out test set was separated by Bemis–Murcko scaffold. In the original saved model runs, XGBoost

with the combined representation produced the lowest held-out MAE, 0.6045 pIC₅₀, with RMSE 0.7981 and $R^2 = 0.5020$. LightGBM with the combined representation achieved MAE 0.6312 and $R^2 = 0.4712$, while Random Forest with the combined representation achieved MAE 0.6985 and $R^2 = 0.4058$.

A graph-based regression model was subsequently implemented using categorical atom features, 13 edge-feature channels, 14 global molecular descriptors, message passing, mean/max graph pooling, and a multilayer regression head. On the saved scaffold-held-out test partition, the GNN achieved MAE 0.6908, RMSE 0.9093, and $R^2 = 0.3535$. A dedicated clean comparison audit, using 3,108 training molecules, 548 validation molecules, and 915 test molecules, similarly found the combined XGBoost model to outperform the GNN on that fixed partition.

The study therefore demonstrates a complete progression from curated target-level bioactivity data to molecular representations, classical QSAR baselines, scaffold-aware validation, and graph-based modelling. The principal scientific contribution of the present work is not a claim of methodological novelty, but a transparent investigation of how data curation, representation choice, validation strategy, and model family affect AChE potency prediction under a chemically separated evaluation protocol.

Keywords: acetylcholinesterase; AChE; ChEMBL; QSAR; pIC₅₀; Morgan fingerprint; RD-Kit descriptors; scaffold split; XGBoost; LightGBM; Random Forest; graph neural network; cheminformatics.

2 Introduction

2.1 Acetylcholinesterase as a Molecular Target

Acetylcholinesterase (AChE) is a serine hydrolase responsible for the rapid hydrolysis of the neurotransmitter acetylcholine. Its catalytic activity regulates the lifetime of acetylcholine in cholinergic synapses, making the enzyme a central component of cholinergic signalling.

From a drug-discovery perspective, AChE is an established molecular target for modulation of cholinergic signalling [Vecchio et al., 2021]. Consequently, large collections of AChE inhibitor measurements have accumulated across medicinal-chemistry programmes and scientific literature. ChEMBL provides a structured, curated representation of bioactivity data suitable for target-focused molecular modelling [Zdrazil et al., 2024, Mendez et al., 2019].

The computational problem addressed here is narrower than therapeutic prediction. The model attempts to learn the relationship between molecular structure and a curated quantitative AChE inhibitory potency label. It does not establish cellular efficacy, pharmacokinetics, toxicity, or clinical benefit.

2.2 Structure–Activity Relationships

Quantitative structure–activity relationship (QSAR) modelling rests on the premise that molecular structure contains information relevant to biological activity. In a medicinal-chemistry dataset, however, this relationship is neither perfectly smooth nor necessarily global. Similar compounds can display substantially different measured activities, while compounds from distinct

structural families can occasionally exhibit comparable potency.

This creates two linked problems.

First, the **molecular representation problem** asks which aspects of chemical structure are made available to the model. A circular fingerprint encodes local substructural environments, whereas physicochemical descriptors encode lower-dimensional global properties. These representations therefore provide different views of the same molecule.

Second, the **generalization problem** asks whether a model predicts genuinely new chemical structures or mainly interpolates among close analogues represented in the training data. Scaffold-aware splitting is commonly used to make this distinction more explicit by assigning compounds sharing a Bemis–Murcko scaffold to the same partition [Bemis and Murcko, 1996].

2.3 From an Educational Workflow to a Structured Investigation

The project began as a reproduction-oriented QSAR workflow and was progressively expanded into a more controlled molecular machine-learning investigation. The final repository contains explicit data-ingestion, cleaning, feature-generation, modelling, validation, audit, and graph-learning components.

The methodological progression can be summarized as:

ChEMBL retrieval → data curation → molecular representations → classical ML → scaffold-aware validation → audit → GNN comparison.

The important change is that model performance is not treated as an isolated number. The project explicitly records the data transformations and structural partitioning that determine what the resulting number means.

2.4 Research Problem

The research problem can be stated as follows:

Given curated molecular structures and experimentally measured AChE IC₅₀ values, how accurately can machine-learning models predict pIC₅₀ under a scaffold-separated evaluation protocol, and how does performance change with molecular representation and model family?

2.5 Research Questions

1. How does the curated AChE activity dataset change as each quality-control stage is applied?
2. How informative are Morgan fingerprints relative to the selected physicochemical descriptors?
3. Does combining local structural fingerprints with global descriptors improve held-out potency prediction?
4. How do Random Forest, XGBoost, and LightGBM compare under the same feature configurations?

5. What does scaffold-separated evaluation reveal about generalization to unseen structural groups?
6. Does the implemented graph representation provide an advantage over the engineered-feature baselines on the saved test partition?

2.6 Working hypotheses

The experimental design was built around four operational hypotheses. First, Morgan fingerprints should provide useful predictive information because they encode local circular structural environments relevant to analogue relationships. Second, the 14 global descriptors should add information not represented explicitly by the fingerprint, so their combination should be competitive with or better than either representation alone. Third, scaffold separation should produce a more chemically demanding estimate than a row-wise random split because close scaffold-related analogues cannot be distributed across train and test. Fourth, an explicit graph representation should be capable of learning useful structure–activity information, but there was no reason to assume in advance that it would outperform strong engineered-feature tree models on a dataset of this size. These are design hypotheses rather than post-hoc claims of success.

3 Dataset and Bioactivity Data

3.1 Target Selection

The repository identifies the target as ChEMBL220, corresponding to acetylcholinesterase, with the raw retrieval saved as `Acetylcholinesterase_Homo_sapiens.csv`. The ingestion code searches ChEMBL target records, selects a target identifier, and retrieves activity records using the target identifier with non-null standard values and `standard_type = IC50`.

The saved audit snapshot contains 8,793 raw activity rows.

3.2 Why the Dataset Is Not Equivalent to a Single Experiment

The data originate from ChEMBL’s structured aggregation of published measurements. Consequently, the raw table is a collection of activity observations rather than a single homogeneous experimental campaign. Each measurement can carry its own assay, source, unit, relation, and molecular identity metadata.

This matters for interpretation. A regression model trained on such data estimates a relationship between molecular structure and the curated activity label as represented in the database. It should not be described as learning an invariant physical constant of AChE binding.

3.3 Raw Dataset Composition

The audit reports the following raw standard-unit distribution:

Table 1: Standard-unit distribution in the saved raw AChE retrieval snapshot.

Original unit	Records	Fraction (%)
nM	8,702	98.965
$\mu\text{g mL}^{-1}$	83	0.944
10'3pM	1	0.011
10'6pM	2	0.023
10'5pM	3	0.034
10^{-4}microM	2	0.023

The current pipeline supports nM and the three encoded pM representations shown above. The unsupported unit records are removed during the supported-unit stage.

3.4 Activity Endpoint and Target Transformation

For an IC_{50} measurement expressed in molar units,

$$p\text{IC}_{50} = -\log_{10}(\text{IC}_{50}[\text{M}]).$$

For example, an IC_{50} of 100 nM corresponds to 10^{-7} M and therefore $p\text{IC}_{50} = 7$.

The saved final cohort has:

- $N = 4571$ molecular records;
- $p\text{IC}_{50}$ minimum = 4.92082;
- $p\text{IC}_{50}$ maximum = 10.96059;
- mean = 6.51519;
- median = 6.30103;
- standard deviation = 1.12667.

The normalized IC_{50} range is 0.01095–10,000 nM.

4 Data Curation

4.1 Curation Philosophy

The curation pipeline is one of the most consequential components of the project because the model cannot distinguish experimental inconsistency from a genuine structure–activity relationship if both are encoded identically in the target variable.

The implemented workflow is:

raw records $\rightarrow$ validity filtering $\rightarrow$ salt/SMILES cleaning
 $\rightarrow$ InChIKey identity $\rightarrow$ potential-duplicate filtering
 $\rightarrow$ unit normalization $\rightarrow$ pIC_{50} $\rightarrow$ replicate QC
 $\rightarrow$ exact deduplication $\rightarrow$ relation filter $\rightarrow$ activity cutoff.

4.2 Curation Audit

Table 2: Saved curation audit from 8,793 raw activity records to 4,571 final molecular records.

Stage	Before	After	Removed	Removed (%)
Raw retrieval	8793	8793	0	0.00
Initial validity/null handling	8793	8072	721	8.20
Salt/SMILES cleaning	8072	8072	0	0.00
InChIKey generation	8072	8072	0	0.00
Potential-duplicate removal	8072	7248	824	10.21
Supported-unit filter	7248	7185	63	0.87
pIC_{50} conversion	7185	7185	0	0.00
Group MAD filter	7185	7110	75	1.04
Exact duplicate removal	7110	6387	723	10.17
Relation = filter	6387	5695	692	10.83
$IC_{50} \leq 10,000$ nM	5695	4571	1124	19.74

The final dataset therefore contains approximately 52.0% of the raw retrieved activity rows. This is not necessarily a weakness: a reduction in row count can be scientifically appropriate when it removes incompatible units, uncertain measurements, replicate inconsistency, or non-equality relations. The important requirement is that every reduction is traceable.

4.3 Structure Standardization

The implementation parses canonical SMILES with RDKit and applies a `SaltRemover`. ^[C1] The cleaned structure is converted to a canonical SMILES representation and then to an InChIKey. The audit found 6,471 unique cleaned SMILES and 6,471 unique InChIKeys at the identity-audit stage, with no multi-mapping between the two representations.

4.4 Replicate Aggregation

Measurements sharing an InChIKey are grouped. For each group, the median pIC_{50} is computed. The absolute deviations from that median are then used to compute a group median absolute deviation (MAD).

The current threshold is:

$$MAD \leq 1.0. \supset [C2]$$

The audit reports 6,421 replicate groups after unit normalization. Thirty-four groups exceed the MAD threshold, corresponding to 74 measurements.

After the MAD filter, the group median is assigned as the pIC_{50} value and exact duplicate rows are removed.

4.5 Equality Relation

The final cohort retains only measurements with:

$$\text{standard_relation} = "="$$

The audit records 692 rows removed at this stage, with 892 raw records carrying a different or null relation before downstream resolution.

This is a deliberate restriction: an exact regression target is easier to interpret when the retained measurement is an equality rather than a censored inequality.

4.6 Activity Cutoff

The final activity cutoff is implemented as: [\[C3\]](#)

$$IC_{50} \leq 10,000 \text{ nM.}$$

A dedicated audit compared the current implementation's `standard_value <= 10000` criterion with the normalized criterion `IC50_nM <= 10000`. Among post-relation candidates, both retained exactly 4,571 records and differed on zero records.

Thus, **the saved final dataset is unchanged by replacing the current expression with an explicitly normalized-unit expression for this snapshot**. Nevertheless, the scientifically clearer implementation is to express the threshold directly in the normalized concentration column.

Curation audit conclusion

The repository contains unusually strong evidence for the data-processing history: stage counts, unit distributions, duplicate analyses, structure warnings, target statistics, scaffold assignments, and leakage checks are saved. This allows the report to describe the final dataset as a defined cohort rather than as an unexplained CSV.

4.7 Structure Sanity Warnings

The audit found:

- 1 RDKit parse failure in the structure-sanity audit;
- 90 disconnected structures after salt-removal processing;
- 0 zero-atom structures;

- 0 unusual-element rows;
- final molecular identities remained internally consistent.

The audit did not remove the 90 disconnected structures. They therefore remain part of the dataset used by downstream feature generation unless separately excluded by another stage.

Limitation requiring explicit discussion

The presence of 90 disconnected structures means that “salt removal” should not be described as proving that every final molecule is a single connected chemical entity. The report should state exactly what the code did and retain the audit warning. A future curation pass should determine whether these disconnected structures are legitimate multi-component molecular records or residual salts/mixtures.

5 Molecular Representation

5.1 Representation as a Scientific Choice

A molecular machine-learning model does not receive “a molecule” in the abstract. It receives a numerical representation generated from that molecule. Therefore, representation choice defines which structural information is accessible to the learning algorithm.

This study compares:

1. Morgan circular fingerprints;
2. a 14-dimensional RDKit descriptor vector;
3. their concatenation.

5.2 Morgan Fingerprints

The implemented fingerprint is generated with RDKit’s Morgan fingerprint generator: [\[C4\]](#)

$$\text{radius} = 2, \quad \text{fpSize} = 1024.$$

A radius of 2 corresponds to circular atom environments extending through local graph neighborhoods of two bond steps. Morgan/ECFP-style fingerprints were developed specifically for structure–activity modelling and encode local circular environments [Rogers and Hahn, 2010]. The resulting representation is a fixed-length 1,024-bit vector.

The audit confirms:

- 4,571 molecules;
- 1,024 fingerprint columns;
- no missing or non-finite values;
- 4,092 unique fingerprint vectors;

- 479 duplicate-excess fingerprint rows;
- mean active bits ≈ 50.46 per molecule.

The 479 duplicate-fingerprint excess is not evidence that the molecules are identical. A fixed-length hashed fingerprint can map different structures to the same bit vector, and chemically different molecules can therefore share the same fingerprint representation.

5.3 RDKit Descriptor Representation

The implemented descriptor set contains exactly 14 variables:

Table 3: Exact RDKit descriptor set used by the repository.

Descriptor	Description
MolWt	Molecular weight
MolLogP	Calculated lipophilicity descriptor
NumHDonors	Hydrogen-bond donor count
NumHAcceptors	Hydrogen-bond acceptor count
TPSA	Topological polar surface area
MaxPartialCharge	Maximum Gasteiger partial charge
MinPartialCharge	Minimum Gasteiger partial charge
NumHeteroatoms	Heteroatom count
NumRotatableBonds	Rotatable-bond count
FractionCSP3	Fraction of carbon atoms that are sp^3
NumAromaticRings	Aromatic ring count
RingCount	Ring count
NumAliphaticNitrogens	Non-aromatic nitrogen count
NumFormalCharge	Formal charge

The feature audit found 29 missing values for both maximum and minimum partial charge. The other descriptors have no missing values in the audit snapshot.

The descriptor set was deliberately small and chemically interpretable. Morgan fingerprints encode discrete local graph environments, but they do not expose continuous global quantities such as molecular size, lipophilicity, polar surface area, hydrogen-bonding capacity, flexibility, ring composition, or formal charge as explicit scalar variables. The selected 14 descriptors therefore provide a compact complementary channel rather than a second high-dimensional representation. Partial-charge extrema were included because they provide a simple electrostatic summary; the remaining variables cover size, polarity, hydrogen-bonding, flexibility, ring topology, carbon hybridization, nitrogen class, and charge. The choice is an engineering design decision intended to balance complementary information against unnecessary descriptor proliferation; the project does not contain an ablation study that isolates the causal contribution of each individual descriptor.

5.4 Combined Representation

The combined representation contains 1,038 explicit model inputs:

$$1024 \text{ Morgan bits} + 14 \text{ RDKit descriptors} = 1038.$$

This is important because the original classical metric CSVs record larger “n_features” values. Those values were generated from the pre-selection data frame shape rather than the actual model matrix. The dedicated GNN/classical audit correctly records 1,038 combined model features.

Implementation/reporting correction

The original metric files report 1,036 Morgan features, 15 descriptor features, and 1,050 combined features. Those numbers are not the actual feature counts presented to the estimators. The modelling code explicitly selects 1,024 Morgan columns and 14 descriptor columns. The extra columns arise because `_save_run_outputs()` records `X_train.shape[1]` before applying the feature whitelist. The report must therefore use 1,024, 14, and 1,038 as the true model input dimensions for the original classical experiments.

6 Scaffold-Aware Partitioning

6.1 Motivation

The repository deliberately moves beyond a purely random row-wise split. Scaffold frameworks provide a reproducible way to group compounds by their central molecular framework [Bemis and Murcko, 1996]. Molecules in medicinal-chemistry datasets often occur in analogue series. If related compounds are distributed across training and test sets, a model can exploit close structural similarity without demonstrating strong extrapolation to a new chemical series.

A scaffold split groups molecules according to Bemis–Murcko scaffolds and keeps each scaffold within a single partition. This is a standard strategy for making chemical generalization more stringent [Bemis and Murcko, 1996].

6.2 Implemented Allocation

The code:

1. calculates a Bemis–Murcko scaffold;
2. converts the scaffold representation to an InChIKey;
3. groups molecules by scaffold identity;
4. sorts scaffold groups by descending group size;
5. appends complete groups to training while the training count is below 80%;

6. assigns all remaining groups to test.

The saved audit confirms:

- training: 3,656 molecules;
- test: 915 molecules;
- training fraction: 79.9825%;
- test fraction: 20.0175%;
- training scaffolds: 1,191;
- test scaffolds: 915;
- scaffold overlap: zero;
- molecule overlap by InChIKey: zero;
- molecule overlap by cleaned SMILES: zero.

6.3 A Critical Property of the Current Split

The split is scaffold-separated, but its allocation rule is not balanced across scaffold sizes. Because scaffold groups are sorted from largest to smallest and training is filled first, the resulting test set consists entirely of singleton scaffolds in the saved audit:

$$\text{test scaffold mean size} = 1.0, \quad \text{test scaffold median size} = 1.0.$$

The largest training scaffold contains 73 molecules, whereas the largest test scaffold contains one molecule.

This is a major interpretive point.

The test set is legitimately **scaffold-disjoint**, but it should not be described as a representative random sample of scaffold groups. It is a chemically separated test set with a strong group-size selection effect induced by the allocation algorithm.

Interpretation boundary

The scaffold split is leakage-safe and chemically separated, but its deterministic size-ordered allocation produces a test set made entirely of singleton scaffolds. The reported metrics therefore characterize this particular partition. The report does not treat them as universal estimates across AChE chemical space.

6.4 Target Distribution Across the Saved Split

The original 3,656/915 split has:

Table 4: pIC₅₀ distribution in the saved original scaffold split.

Partition	<i>N</i>	Mean	Median	SD	Range
Train	3656	6.48356	6.26761	1.12338	4.92082–10.96059
Test	915	6.64156	6.52288	1.13152	4.97879–9.92082

The test median is somewhat higher than the training median. This should be reported descriptively, not interpreted as evidence of a mechanistic potency difference.

7 Classical Machine Learning

7.1 Experimental Matrix

The repository evaluates three model families across three representations:

Table 5: Classical modelling matrix.

Model	Morgan	Descriptors	Combined
Random Forest	Yes	Yes	Yes
XGBoost	Yes	Yes	Yes
LightGBM	Yes	Yes	Yes

The comparison is therefore factorial in spirit: representation and model family can both be examined. [\[C5\]](#)

7.2 Random Forest

The implementation uses `RandomForestRegressor` as the bagged-tree baseline, following the general Random Forest framework [Breiman, 2001], with hyperparameters selected through Optuna. The search includes tree count, depth, minimum split/leaf sizes, feature subsampling, bootstrap selection, and cost-complexity pruning.

The final saved original run for the combined representation used a model selected through the repository’s five-fold scaffold-grouped CV objective.

7.3 XGBoost

The XGBoost implementation varies tree count, depth, learning rate, subsampling, feature subsampling, and L1/L2 regularization. The combined representation is the strongest original XGBoost configuration.

7.4 LightGBM

LightGBM is similarly optimized; LightGBM is a gradient-boosted decision-tree implementation designed for efficient high-dimensional learning [Ke et al., 2017], over tree count, depth, leaf

count, learning rate, subsampling, feature subsampling, and regularization.

The best saved LightGBM combined trial achieved a CV objective corresponding to MAE 0.63128 pIC₅₀.

7.5 Optuna Objective

The modelling objective uses: [\[C6\]](#)

```
scoring = neg_mean_absolute_error.
```

Because scikit-learn represents loss metrics as negative scores for maximization-compatible cross-validation APIs, the saved values are negative. The best value is therefore the value closest to zero.

For reporting, the scientifically readable quantity is:

$$\text{CV MAE} = -\text{cv_neg_mae}.$$

Correct interpretation

The negative CV values in the CSVs are not negative prediction errors. They are scikit-learn’s “negative MAE” scoring convention. The report converts them to positive MAE when discussing model performance.

8 Original Classical Results

8.1 Held-Out Performance

Table 6: Original saved classical-model results on the 3,656/915 scaffold split. Feature counts are corrected to the actual estimator input dimensions.

Model	Representation	N_f	MAE	RMSE	R^2	Pearson r	Spearman ρ
RF	Morgan	1024	0.7053	0.8840	0.3889	0.6325	0.6372
RF	Descriptors	14	0.7186	0.8983	0.3690	0.6155	0.5949
RF	Combined	1038	0.6985	0.8718	0.4058	0.6476	0.6446
XGB	Morgan	1024	0.6231	0.8240	0.4691	0.6905	0.6765
XGB	Descriptors	14	0.6613	0.8568	0.4259	0.6533	0.6291
XGB	Combined	1038	0.6045	0.7981	0.5020	0.7109	0.6935
LGBM	Morgan	1024	0.6530	0.8560	0.4271	0.6625	0.6558
LGBM	Descriptors	14	0.6780	0.8775	0.3979	0.6340	0.6082
LGBM	Combined	1038	0.6312	0.8223	0.4712	0.6885	0.6746

The strongest original saved classical model is XGBoost with the combined representation:

$$\text{MAE} = 0.60447, \quad \text{RMSE} = 0.79808, \quad R^2 = 0.50198.$$

An MAE of 0.6045 pIC₅₀ corresponds to a multiplicative factor of approximately

$$10^{0.6045} \approx 4.02.$$

This is a useful chemical-scale interpretation of the log-space error, but it should not be described as “the model is exactly four-fold wrong” for every compound. MAE is an average absolute error in logarithmic potency space.

8.2 Representation Effects

Within each model family, the combined representation improves over both individual representations in the saved test results.

For XGBoost:

$$\text{MAE}_{\text{Morgan}} = 0.6231,$$

$$\text{MAE}_{\text{Descriptors}} = 0.6613,$$

$$\text{MAE}_{\text{Combined}} = 0.6045.$$

Thus, the combined representation improves the XGBoost test MAE relative to Morgan alone by approximately 0.0187 pIC₅₀ and relative to descriptors alone by approximately 0.0568 pIC₅₀.

The result is consistent with the intended complementary-role hypothesis: local structural patterns and global physicochemical descriptors contain overlapping but non-identical information.

However, this is an empirical result for this dataset and split. It does not establish that combined features will universally outperform fingerprints alone.

8.3 Model Effects

For the combined representation:

$$\text{MAE}_{\text{RF}} = 0.6985, \quad \text{MAE}_{\text{XGB}} = 0.6045, \quad \text{MAE}_{\text{LGBM}} = 0.6312.$$

XGBoost is therefore the strongest of the three saved combined-feature models on this test partition.

The difference between XGBoost and LightGBM is approximately 0.0267 pIC₅₀, while the difference between XGBoost and Random Forest is approximately 0.0940 pIC₅₀.

These differences are descriptive. No repeated-split statistical test is present in the repository that would establish the stability of the ranking.

9 Hyperparameter Optimization

9.1 What Was Actually Run

The saved original Optuna CSVs contain five completed trials for the principal model runs. Therefore, the report should not claim that the final published metrics were selected from 100 trials merely because the function default permits a larger number.

For example, the saved combined-feature XGBoost trials contain five completed configurations, with the best validation objective corresponding to:

$$\text{CV MAE} = 0.59735.$$

The saved combined-feature LightGBM trials also contain five completed configurations, with best:

$$\text{CV MAE} = 0.63128.$$

The saved combined-feature Random Forest trials contain five completed configurations, with best:

$$\text{CV MAE} = 0.68116.$$

9.2 Saved Best Parameters

Table 7: Best saved parameters for the original combined-feature runs.

Model	Best parameters
Random Forest	$n_{\text{estimators}} = 760$, $max_depth = 34$, $min_samples_leaf = 5$, $min_samples_split = 19$, $max_features = \text{sqrt}$, $bootstrap = \text{True}$, $ccp_alpha = 2.149 \times 10^{-7}$
XGBoost	$n_{\text{estimators}} = 1181$, $max_depth = 10$, learning rate = 0.02531, subsample = 0.67327, column subsampling = 0.67588, $\alpha = 2.04 \times 10^{-7}$, $\lambda = 5.675$
LightGBM	$n_{\text{estimators}} = 539$, $max_depth = 11$, $num_leaves = 177$, learning rate = 0.03227, subsample = 0.75359, column subsampling = 0.76205, $\alpha = 2.249$, $\lambda = 0.001388$

The search spaces were chosen to span the principal capacity and regularization controls of each tree learner without making the exploratory stage computationally prohibitive. Random Forest varies ensemble size, depth, leaf/split constraints, feature subsampling, bootstrap behaviour, and pruning. XGBoost and LightGBM vary tree count, depth, learning rate, row/column subsampling, and L1/L2 regularization, with LightGBM additionally varying leaf count. The ranges are broad enough to include shallow and moderately deep trees, slower and faster boosting schedules, and weak-to-strong regularization. They are not exhaustive global searches. The

saved original experiments contain five completed trials per principal model/representation, while the later clean comparison uses eight trials and three-fold scaffold-grouped CV; those two experiment generations are kept separate.

10 Graph Neural Network

10.1 Motivation

The GNN stage changes the representation. Message-passing neural networks provide a general framework for learning molecular representations directly from graph structure [Gilmer et al., 2017], from a fixed-length engineered feature vector toward an explicit molecular graph. A molecule is represented as

$$G = (V, E),$$

where V is the set of atoms and E is the set of bonds.

The repository’s graph model is not a generic “black-box GNN”; it explicitly encodes atom categories, bond features, and global molecular descriptors.

10.2 Atom Features

Seven categorical atom features are implemented:

1. atomic number;
2. degree;
3. formal charge;
4. hybridization;
5. aromaticity;
6. hydrogen count;
7. ring membership.

The categorical values are clipped to predefined ranges and represented through learned embeddings.

10.3 Edge Features

The implementation contains 13 edge channels:

- four bond-type indicators;
- conjugation;
- aromaticity;

- ring membership;
- six stereo indicators.

Each chemical bond is represented in both directions, producing directed message edges.

Important documentation correction

The code implements 13 edge-feature channels. Any older project note describing the GNN as using exactly three edge inputs is inconsistent with the current implementation and should not appear in the final report.

10.4 Global Descriptors

The GNN also receives the same 14-dimensional global descriptor vector used by the classical descriptor experiments.

Thus the model is best described as a **graph-plus-descriptor molecular regressor**, not a graph-only model.

10.5 Message Passing

The graph network begins with independent embeddings for the seven categorical atom features. [C7] These embeddings are concatenated and projected to a hidden dimension of 128.

Three message-passing steps are then applied. For each directed edge, the source node representation is concatenated with the 13-dimensional edge feature vector and passed through a linear transformation followed by a ReLU activation. Messages are aggregated at the destination node by summation and normalized by destination degree.

A GRU cell updates the node representation, followed by residual addition, dropout, and layer normalization.

This can be represented schematically as:

$$h_v^{(k+1)} = \text{LN} \left(h_v^{(k)} + \text{Dropout} \left[\text{GRU} \left(\frac{1}{d_v} \sum_{u \in \mathcal{N}(v)} \phi_k(h_u^{(k)}, e_{uv}), h_v^{(k)} \right) \right] \right).$$

The equation describes the implemented computational structure; it should not be interpreted as a claim that the architecture has been theoretically optimized for AChE.

10.6 Graph Readout

After message passing, mean and max pooling are calculated over atom representations for each molecule. These pooled graph representations are concatenated with the standardized 14-dimensional descriptor vector and passed to a multilayer regression head.

The final architecture therefore combines:

atom embeddings $\rightarrow$ message passing $\rightarrow$ mean/max pooling
 $\rightarrow$ 14 global descriptors $\rightarrow$ MLP $\rightarrow p\hat{IC}_{50}$.

10.7 Training

The saved GNN run uses: [\[C8\]](#)

- hidden size = 128;
- message-passing steps = 3;
- batch size = 64;
- learning rate = 0.001;
- weight decay = 0.0001;
- Smooth L1 loss;
- AdamW optimizer;
- gradient clipping at 5.0;
- early stopping patience = 20 epochs;
- seed = 50;
- CPU execution.

Molecular descriptors are standardized from the training partition. Missing partial-charge values are imputed from training-derived means.

The target pIC_{50} is also standardized during training and transformed back to the original scale for evaluation.

10.8 GNN Result

The saved GNN result is:

$$\text{MAE} = 0.69078, \quad \text{RMSE} = 0.90931, \quad R^2 = 0.35349, \quad r_{\text{Pearson}} = 0.62255.$$

The best validation MAE was 0.65072 at epoch 48, and training stopped after epoch 68 because the validation criterion did not improve for the patience window.

The GNN therefore did not outperform the best classical model in this experiment.

11 Clean GNN–Classical Comparison Audit

11.1 Purpose

The repository contains a dedicated audit specifically designed to compare the GNN with classical baselines on matched partitions. This audit is particularly important because it removes

ambiguity around the validation partition and feature-count reporting.

The clean comparison uses:

$$N_{\text{train}} = 3108, \quad N_{\text{validation}} = 548, \quad N_{\text{test}} = 915.$$

The 915 test molecules are the same saved scaffold-held-out test partition, while 548 molecules are reserved from the original training portion for validation.

11.2 Clean Comparison Results

Table 8: Clean GNN/classical comparison audit.

Model	Features	N_f	CV MAE	Test MAE	Test RMSE	R^2
XGBoost	Combined	1038	0.6479	0.6352	0.8347	0.4553
XGBoost	Morgan	1024	0.6569	0.6510	0.8519	0.4326
RF	Combined	1038	0.6875	0.6764	0.8507	0.4341
RF	Morgan	1024	0.6948	0.6852	0.8631	0.4175
GNN	Graph + descriptors	34	—	0.6908	0.9093	0.3535
XGBoost	Descriptors	14	0.7276	0.6969	0.8892	0.3818
RF	Descriptors	14	0.7366	0.7169	0.8978	0.3698

On this fixed comparison, the combined XGBoost model has a test MAE 0.0556 pIC₅₀ lower than the GNN.

This result does **not** establish that GNNs are intrinsically inferior for AChE potency prediction. It establishes only that this implemented GNN architecture, training procedure, representation, and split did not outperform the matched classical baselines on this partition.

11.3 Why the GNN Result Is Scientifically Useful

A negative result is informative when the comparison is controlled. The GNN stage demonstrates that replacing engineered features with an explicit graph architecture does not automatically improve prediction.

Several explanations remain possible:

1. the classical representations may already capture much of the predictive signal available in the dataset;
2. the dataset size may not be sufficient for the chosen GNN capacity;
3. the graph architecture may not encode the most relevant interaction information;
4. the model may require different regularization or hyperparameter tuning;
5. the 14 global descriptors already provide strong low-dimensional information;
6. the unusual scaffold-test construction may influence model ranking.

These are hypotheses, not conclusions established by the current experiments.

12 Results and Interpretation

12.1 What the Best Model Actually Demonstrates

The strongest saved classical result is:

$$\text{XGBoost} + \text{Morgan} + \text{RDKit} = 0.6045 \text{ pIC}_{50} \text{ MAE}$$

on the original 915-molecule scaffold-held-out test partition.

This means that, under the saved split and evaluation protocol, the model’s average absolute deviation from measured pIC_{50} is approximately 0.60 log units.

The R^2 value of 0.502 indicates that approximately half of the variance relative to the test-set mean is explained by the predictions under the conventional coefficient-of-determination definition. It does not mean that the model has “50.2% accuracy.”

12.2 Correlation Versus Error

Pearson correlation for the best original model is 0.7109 and Spearman correlation is 0.6935.

The distinction is useful:

- Pearson correlation assesses linear association;
- Spearman correlation assesses rank association;
- MAE and RMSE quantify absolute prediction error;
- R^2 measures variance explained relative to a baseline mean predictor.

A model can therefore have a respectable correlation while still making sizeable absolute potency errors.

12.3 Why RMSE Matters

For the best XGBoost model:

$$\text{MAE} = 0.6045, \quad \text{RMSE} = 0.7981.$$

Because RMSE exceeds MAE, the error distribution contains observations with larger deviations than the average absolute error. The current repository contains prediction and residual files that can be used to identify these compounds.

The saved prediction files support molecule-level residual inspection, but the present evidence does not include a formally completed chemical nearest-neighbour or activity-cliff analysis. The appropriate interpretation is therefore diagnostic rather than mechanistic. The model diagnostics show substantial residual tails for every classical configuration, consistent with a subset of compounds being considerably harder to predict than the central portion of the test distribution. Because the test set is composed of singleton scaffolds, large residuals cannot automatically be

attributed to scaffold novelty: every test molecule is scaffold-disjoint, but the repository does not provide a quantified similarity-to-training analysis. Likewise, assay heterogeneity is plausible because ChEMBL aggregates measurements across assays and sources, but the present model does not establish that assay metadata explains individual residuals. These distinctions are retained explicitly to avoid turning residual patterns into unsupported mechanistic explanations.

13 Error Analysis

13.1 Residual Definition

For molecule i ,

$$e_i = y_i - \hat{y}_i.$$

Positive residuals indicate underprediction of potency when potency is represented by pIC_{50} ; negative residuals indicate overprediction.

13.2 Available Diagnostic Outputs

The repository saves:

- per-model prediction CSVs;
- residual values;
- prediction-versus-actual diagnostic plots;
- residual histograms;
- combined prediction tables;
- model-comparison metrics.

This provides a direct route to a molecule-level error analysis rather than relying only on aggregate metrics.

13.3 Activity Cliffs

Activity cliffs are chemically similar molecules with substantially different activity. They are important because they violate a simple smoothness assumption underlying many QSAR models.

The current repository does not yet provide a formal activity-cliff analysis sufficient to claim that specific cliffs explain the model’s errors. Therefore, the report treats activity cliffs as an analysis direction rather than as a demonstrated causal explanation.

A formal activity-cliff analysis is not part of the saved project outputs and is therefore not claimed here. The concept remains relevant because a fingerprint-based model can be challenged by chemically close molecules with sharply different measured potency. A future analysis should define a similarity metric and threshold before inspecting the residuals, then test whether

high-similarity/high-potency-difference pairs are enriched among the largest errors. Until that analysis exists, activity cliffs are a plausible explanation for some errors, not a demonstrated explanation of this model's errors.

13.4 Applicability Domain

A single global MAE does not tell us whether every compound is equally predictable. A model can perform well on dense regions of chemical space while failing on structurally isolated molecules. The current repository does not contain a finalized applicability-domain method. Therefore, no quantitative applicability-domain claim should be included in the abstract or conclusion.

Potential future analyses include nearest-neighbour Tanimoto similarity, distance-to-training-space measures, conformal prediction, or uncertainty estimates.

14 Scientific Interpretation of Molecular Representations

14.1 Morgan Fingerprints: What They Give the Model

Morgan fingerprints expose local circular structural environments. They can therefore distinguish molecules according to patterns of atom connectivity and nearby substituent context.

The representation is particularly useful for analogue-series learning because chemically relevant substructures can recur across compounds.

However, the representation is hashed into 1,024 bits. Distinct substructures can therefore collide into the same bit position, and the fingerprint does not directly encode continuous physicochemical quantities such as molecular weight or logP.

14.2 RDKit Descriptors: What They Add

The descriptor vector introduces explicit global properties.

For example:

- molecular weight captures overall molecular size;
- MolLogP represents calculated lipophilicity;
- H-bond donor/acceptor counts summarize hydrogen-bonding capacity;
- TPSA summarizes polar surface contribution;
- rotatable-bond count reflects conformational flexibility;
- aromatic and total ring counts provide coarse structural topology;
- charge descriptors add electrostatic information.

These quantities are not equivalent to binding affinity. They are structural proxies that may correlate with features influencing molecular recognition, solubility, permeability, or conformational behaviour.

14.3 Why the Combined Representation Helps in This Dataset

The saved results show that combining Morgan fingerprints with the 14 descriptors improves every classical model family relative to the corresponding descriptor-only configuration and improves each model family relative to its descriptor-only result.

For XGBoost, the combined representation is the strongest:

$$0.6045 < 0.6231 < 0.6613.$$

The data therefore support the statement that the combined feature space was more predictive for this AChE dataset under the saved test protocol than either representation alone.

The data do not prove which individual descriptor caused the improvement.

15 Reproducibility

15.1 Repository Organization

The repository contains source modules for:

- data ingestion;
- data cleaning;
- data engineering;
- validation/auditing;
- classical model building;
- GNN construction and training;
- GNN/classical comparison.

Saved data and outputs include raw and cleaned CSV snapshots, model metrics, predictions, Optuna trials, audit tables, model checkpoints, and diagnostic figures.

15.2 Recorded Software Environment

The saved audit configuration reports:

Table 9: Environment recorded in the repository audit snapshot.

Python	3.12.3
RDKit	2026.03.4
Pandas	2.3.3
NumPy	2.4.3
scikit-learn	1.9.0
SciPy	1.17.1

The audit also records a Git commit hash and data SHA-256 values, providing an explicit provenance trail for the saved dataset snapshot.

15.3 Seeds

The principal model code uses seed 50 for Random Forest and the Optuna TPE sampler. The GNN also uses seed 50 for Python, NumPy, and PyTorch.

A single seed does not quantify training variability. Repeated-seed experiments would be required to report mean $\pm$ standard deviation of model performance.

16 Implementation Audit and Required Corrections

This section is intentionally included so that the final report does not overstate the cleanliness of the current repository.

16.1 Issue 1: Feature Count in Original Metric CSVs

Status: confirmed reporting bug; does not change predictions.

The original metrics use:

$$\text{n_features} = X_{\text{train}}.\text{shape}[1]$$

before the explicitly selected model columns are passed to the estimator.

Therefore:

$$1024 \neq 1036, \quad 14 \neq 15, \quad 1038 \neq 1050.$$

The underlying model matrices are correctly selected by `_model_columns()`. The error is in the logged metadata, not in the feature matrix used for prediction.

Action: use corrected feature counts in the manuscript and, ideally, regenerate the metric CSVs.

16.2 Issue 2: Scaffold Allocation Bias

Status: confirmed methodological limitation.

The current split sorts scaffolds by descending group size and fills the training set first. The saved test set therefore contains only singleton scaffolds.

This prevents scaffold overlap but introduces a strong test-set composition effect.

Action: do not claim the current test set is a balanced or random sample of scaffolds. For a stronger study, repeat the analysis using randomized or balanced scaffold assignment across multiple seeds.

16.3 Issue 3: Disconnected Structures

Status: audit warning.

Ninety disconnected structures remain after salt removal according to the saved audit. The audit did not remove them.

Action: inspect these records before the final version of the manuscript. Determine whether they represent legitimate multi-component structures, residual salts, mixtures, or other chemically meaningful records.

16.4 Issue 4: Morgan Fingerprint Function Signature

The method `morgan_fingerprinting_s_method()` accepts `radius` and `nBits` arguments but uses the class-level Morgan generator configured as radius 2 and `fpSize` 1024 regardless of those arguments.

The current calls use radius 2 and 1024, so this does not alter the saved experiments. Nevertheless, the function interface is misleading.

Action: either remove the unused parameters or instantiate a generator from the requested parameters.

16.5 Issue 5: Activity Threshold Expression

The current pipeline uses `standard_value <= 10000`. Because the retained supported units are converted with fixed factors, the audit shows that this produces exactly the same 4,571-row cohort as `IC50_nM <= 10000` for the saved snapshot.

Action: prefer the normalized-unit expression in future code because it communicates the scientific threshold directly and is less dependent on the current unit whitelist.

16.6 Issue 6: GNN Edge-Feature Documentation

The current GNN has 13 edge channels, not three.

Action: use the current code as the authoritative architecture description.

16.7 Issue 7: GNN Is Not Graph-Only

The GNN concatenates mean/max graph embeddings with 14 global molecular descriptors.

Action: describe it as a graph-plus-descriptor model.

16.8 Issue 8: Single Fixed Split

Both the original classical results and the clean GNN comparison are based on a fixed scaffold-held-out test partition.

Action: phrase conclusions as “on this held-out partition” and avoid claims of universal superiority or generalization across all AChE chemical space.

17 Limitations

17.1 Dataset Limitations

The dataset aggregates measurements from multiple experimental sources. Assay conditions, protocols, laboratories, and experimental noise may therefore contribute variation that is not encoded in molecular structure.

17.2 Endpoint Limitations

IC₅₀ is an assay-dependent quantity. It is not identical to a thermodynamic binding constant and is not sufficient to establish cellular efficacy or therapeutic activity.

17.3 Representation Limitations

Morgan fingerprints encode local two-dimensional structural environments but do not explicitly encode three-dimensional conformation, protein interactions, solvent effects, or dynamic binding behaviour.

The selected descriptors similarly provide coarse molecular properties rather than a complete physical description of ligand–enzyme recognition.

17.4 Validation Limitations

The current scaffold split is leakage-safe but allocation-biased toward singleton test scaffolds. This makes repeated or independently generated scaffold partitions particularly important.

17.5 Model Limitations

Only three classical model families are benchmarked. The GNN is a single architecture/configuration rather than an exhaustive architecture search.

17.6 Biological Limitations

The prediction target is molecular potency against a target-level activity endpoint. The model does not predict absorption, distribution, metabolism, excretion, toxicity, selectivity, blood–brain-barrier penetration, cellular response, or clinical outcome.

18 Conclusions

This study establishes a reproducible computational pipeline for AChE inhibitory-potency prediction from curated molecular structures.

The principal findings are:

1. The raw AChE activity snapshot contains 8,793 records, which are reduced to 4,571 unique molecular records through explicit curation stages.
2. The curation workflow removes invalid/flagged records, unsupported units, inconsistent replicate groups, duplicate records, non-equality measurements, and measurements above the final activity threshold.
3. The final pIC₅₀ distribution spans 4.92082–10.96059, with median 6.30103.
4. Morgan fingerprints and RDKit descriptors provide complementary molecular representations.
5. The combined representation performs best among the original classical configurations for all three tested model families.
6. XGBoost with combined features produces the strongest original held-out result: MAE 0.6045, RMSE 0.7981, and $R^2 = 0.5020$.
7. The implemented GNN does not outperform the strongest classical baseline on the saved test partition.
8. The repository audit identifies important methodological/reporting issues, especially the size-ordered scaffold allocation and the original feature-count logging error.

The most defensible overall conclusion is therefore:

For the curated ChEMBL220 AChE dataset and the saved scaffold-held-out evaluation protocol, engineered molecular representations combined with gradient-boosted tree learning provide stronger predictive performance than the implemented graph-based model. The result supports the usefulness of complementary local structural and global physicochemical information for this dataset, while the fixed and size-ordered scaffold partition limits the strength of any claim about generalization to broader chemical space.

19 Future Work

19.1 Immediate Methodological Corrections

The project already contains the principal evidence required for this technical report. The remaining items are methodological extensions rather than prerequisites for documenting the completed work:

1. inspect the 90 disconnected structures to determine their chemical status;
2. regenerate the original metric metadata so the stored feature counts agree with the estimator inputs;
3. retain the normalized-unit form of the activity cutoff explicitly in future pipeline revisions;
4. replace the single size-ordered scaffold partition with repeated or better-balanced scaffold assignment if a stronger generalization study is desired;
5. repeat the principal experiments under the revised split if a future study requires uncertainty estimates for the model ranking.

19.2 Chemical Error Analysis

The most scientifically valuable next stage is not simply adding another model. It is understanding *where and why* the current best model fails.

The saved prediction tables allow:

1. ranking compounds by absolute residual;
2. identifying nearest-neighbour similarity to training molecules;
3. comparing errors across scaffold novelty;
4. searching for activity cliffs;
5. evaluating error against physicochemical properties;
6. testing whether the model systematically underpredicts or overpredicts certain chemical regimes.

19.3 Repeated Scaffold Validation

A stronger estimate should report:

$$\text{MAE} = \mu \pm \sigma$$

across repeated scaffold partitions rather than relying on one fixed split.

This would allow the study to distinguish a stable modelling effect from a property of one particular test partition.

19.4 Graph Modelling

The GNN can be developed further by varying:

- message-passing architecture;
- hidden dimension;
- depth;
- pooling;
- edge representation;
- descriptor integration;
- regularization;
- repeated seeds.

The purpose should be to test specific hypotheses rather than simply increase architectural complexity.

19.5 External Validation

A genuinely stronger generalization claim would require an external dataset or a temporally separated evaluation set.

19.6 Mechanistic Extensions

Future work may incorporate three-dimensional molecular information, protein–ligand interaction information, docking-derived features, uncertainty estimation, and scientifically constrained graph learning. These should be treated as future hypotheses rather than retroactively presented as completed components of the present study.

20 Scientific Decision Rationale

This section records the reasoning behind the principal fixed choices so that the report is not merely a transcription of code.

20.1 Why CHEMBL220 and IC₅₀?

The project is explicitly target-specific: CHEMBL220 is the selected acetylcholinesterase target, and IC₅₀ is the quantitative activity endpoint retrieved by the ingestion layer. Restricting the study to one target avoids mixing activity mechanisms from unrelated proteins. IC₅₀ was retained because it is the endpoint actually available at scale in the retrieved target activity table; converting it to pIC₅₀ places potency on a logarithmic molar scale that is more convenient for regression and compresses the wide concentration range.

20.2 Why retain only equality relations?

Values reported as $>$, $<$, or approximate relations are censored or non-exact measurements. Treating them as ordinary point targets would silently convert partial information into false precision. The project therefore retains “=” measurements for the final regression target. This sacrifices some data quantity in exchange for a cleaner numerical interpretation.

20.3 Why median aggregation and MAD = 1.0?

Multiple measurements for the same standardized molecular identity can arise from different assays or records. The median is robust to isolated extreme values and provides a single representative target without privileging one arbitrary record. MAD is then used as a robust within-identity consistency screen. A threshold of 1.0 pIC₅₀ is intentionally permissive: it flags groups with substantial disagreement rather than deleting modest experimental variation. The audit shows that only 34 of 6,421 replicate groups exceed this threshold, so the rule is selective rather than an aggressive pruning mechanism.

20.4 Why 10,000 nM?

The final activity range is restricted to measurements at or below 10,000 nM. This keeps the regression focused on measurable inhibitor potency rather than extremely weak activity. The audit independently tested the raw-value and normalized-nM interpretations for this dataset and found zero differently classified records; therefore the choice does not alter the saved final cohort. Future code should still express the rule directly in normalized nM because that makes the scientific unit explicit.

20.5 Why Morgan radius 2 and 1,024 bits?

Radius 2 gives each atom environment access to a local neighbourhood extending two bond steps, providing a middle ground between very local atom/bond information and increasingly large subgraph environments. A 1,024-bit vector keeps the representation computationally manageable while retaining thousands of possible hashed environments. The project does not claim that radius 2 or 1,024 bits is globally optimal; these are fixed representation settings used consistently across the benchmark.

20.6 Why these 14 descriptors?

The descriptor vector was intentionally kept compact and complementary to the fingerprint. The set spans size, lipophilicity, hydrogen bonding, polarity, charge, heteroatom content, flexibility, carbon hybridization, and ring topology. These quantities give tree models direct access to continuous global molecular properties that a hashed binary fingerprint does not expose as explicit scalar variables.

20.7 Why scaffold separation?

The scientific question is not merely whether the model can interpolate among close analogues. A scaffold split asks how the learned relationship behaves when the central molecular framework is absent from training. Bemis–Murcko scaffolds provide a reproducible two-dimensional framework definition. The implementation groups complete scaffolds so that no scaffold crosses the held-out boundary.

20.8 Why five-fold GroupKFold for the original classical search?

The hyperparameter objective is computed on training data only, and scaffold identifiers are passed as groups. Five folds provide repeated validation partitions while preventing the same scaffold from appearing in both sides of a fold. This is preferable to ordinary K-fold row splitting for a scaffold-generalization question because the latter could place close structural families on both sides of the validation boundary.

20.9 Why Optuna and TPE?

The project uses Optuna with a TPE sampler seeded at 50. Tree models expose several interacting hyperparameters, including depth, ensemble size, learning rate, subsampling and

regularization. A sequential sampler is therefore more efficient for the exploratory search than manually enumerating a large Cartesian grid. The number of completed trials is nevertheless small in the saved original experiments, so the optimization should be understood as targeted tuning rather than exhaustive optimization.

20.10 Why the GNN architecture?

The GNN was designed to expose information that fixed fingerprints encode only indirectly: atom identities and local connectivity, bond type/stereo information, and graph-level pooled structure. Seven atom attributes are embedded independently, then projected to 128 dimensions. Three message-passing steps allow information to propagate beyond immediate neighbours without making the network unnecessarily deep for the molecular sizes in the dataset. Mean and max pooling provide complementary summaries, while the 14 descriptors preserve the same global information available to the classical combined model. Smooth L1 loss reduces sensitivity to large individual errors compared with pure squared loss; AdamW supplies adaptive optimization with decoupled weight decay; gradient clipping at 5.0 limits unstable updates; dropout 0.15 provides moderate regularization; and patience 20 prevents unnecessary training after validation performance has stopped improving. These choices are coherent engineering choices for the available dataset, not individually optimized scientific constants.

20.11 Why CPU and seed 50?

The saved GNN run was executed on CPU with seed 50. The seed is propagated through Python, NumPy and PyTorch in the training script, making the saved experiment deterministic to the extent permitted by the software stack and hardware. CPU execution was sufficient for the completed run and avoids dependence on a particular CUDA environment in the recorded experiment.

20.12 What is deliberately not claimed?

The project does not claim therapeutic efficacy, binding mechanism, prospective drug-discovery utility, universal chemical-space generalization, state-of-the-art performance, or intrinsic inferiority of GNNs. The results are statements about the curated ChEMBL220 dataset and the saved evaluation protocols.

A Supplementary Table: Original Metrics

Table 10: Full original metric records.

Model	Features	MAE	RMSE	R^2	Pearson	Spearman	CV MAE	Train	Test
RF	Morgan	0.7053	0.8840	0.3889	0.6325	0.6372	0.6854	3656	915
RF	Descriptors	0.7186	0.8983	0.3690	0.6155	0.5949	0.7202	3656	915
RF	Combined	0.6985	0.8718	0.4058	0.6476	0.6446	0.6812	3656	915
XGB	Morgan	0.6231	0.8240	0.4691	0.6905	0.6765	0.6042	3656	915
XGB	Descriptors	0.6613	0.8568	0.4259	0.6533	0.6291	0.6993	3656	915
XGB	Combined	0.6045	0.7981	0.5020	0.7109	0.6935	0.5974	3656	915

Model	Features	MAE	RMSE	R^2	Pearson	Spearman	CV MAE	Train	Test
LGBM	Morgan	0.6530	0.8560	0.4271	0.6625	0.6558	0.6347	3656	915
LGBM	Descriptors	0.6780	0.8775	0.3979	0.6340	0.6082	0.7141	3656	915
LGBM	Combined	0.6312	0.8223	0.4712	0.6885	0.6746	0.6313	3656	915

B Supplementary Table: Clean GNN/Classical Audit

Table 11: Clean matched comparison using 3,108 training, 548 validation, and 915 test molecules.

Model	Features	N_f	CV MAE	Test MAE	RMSE	R^2	Pearson
XGB	Combined	1038	0.6479	0.6352	0.8347	0.4553	0.6795
XGB	Morgan	1024	0.6569	0.6510	0.8519	0.4326	0.6648
RF	Combined	1038	0.6875	0.6764	0.8507	0.4341	0.6644
RF	Morgan	1024	0.6948	0.6852	0.8631	0.4175	0.6506
GNN	Graph+Descriptors	34	—	0.6908	0.9093	0.3535	0.6225
XGB	Descriptors	14	0.7276	0.6969	0.8892	0.3818	0.6188
RF	Descriptors	14	0.7366	0.7169	0.8978	0.3698	0.6140

C Diagnostic Figures

The report embeds the figures shipped with the project archive. Each figure is treated as an experimental output rather than decorative material. The main text uses the figures to support interpretation; the appendix provides the complete set of model-specific diagnostics.

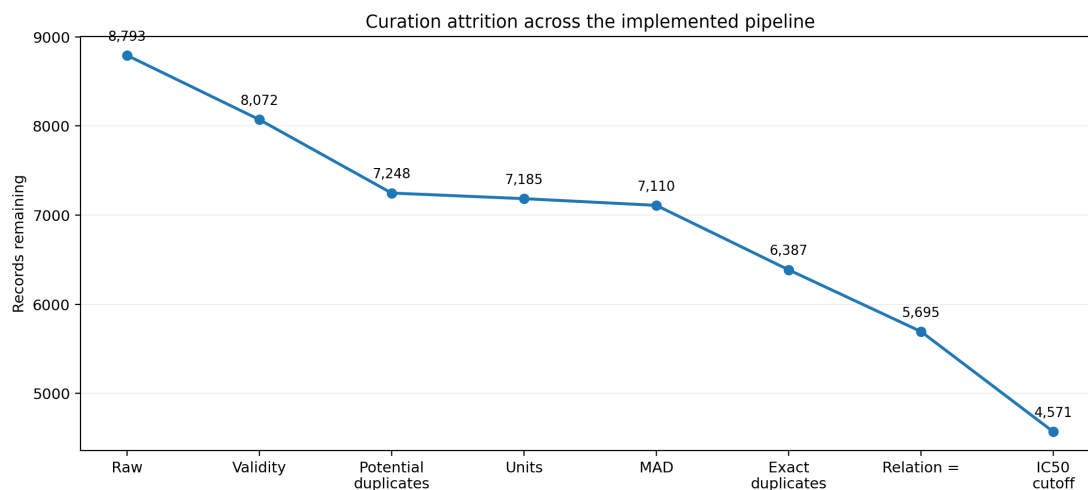


Figure 1: Curation attrition across the implemented pipeline. The largest single reduction occurs at the final activity cutoff, after earlier validity, duplicate, relation and replicate-quality filters.

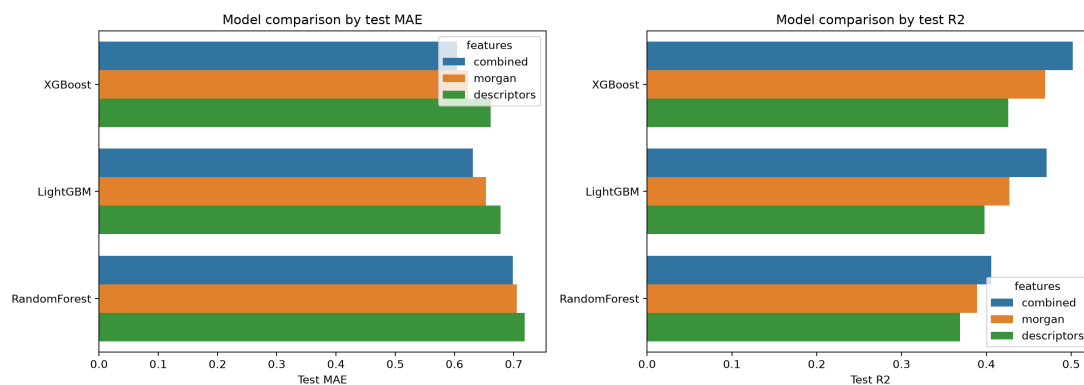


Figure 2: Saved original classical-model comparison by held-out test MAE and R^2 . Combined features are consistently competitive across the three tree learners, with XGBoost + combined features giving the strongest saved result.

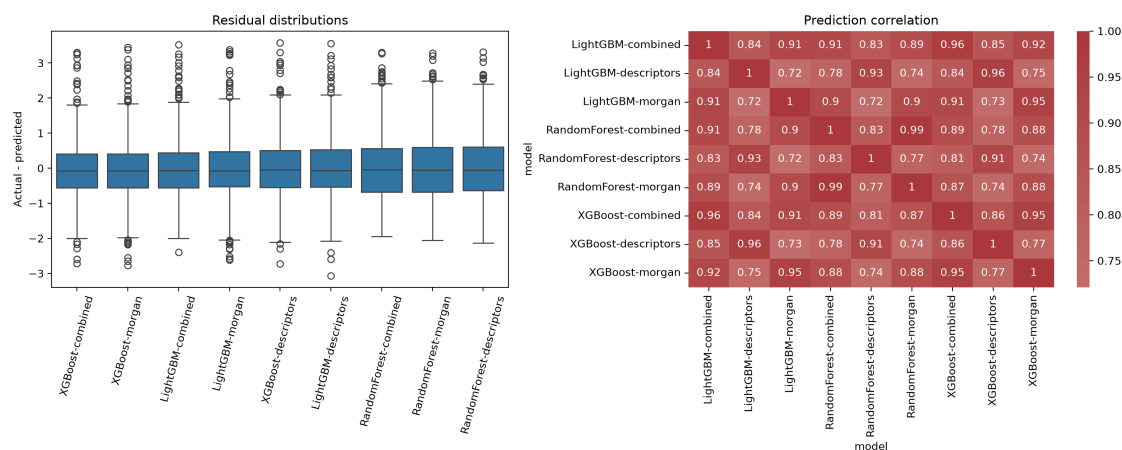


Figure 3: Project-level diagnostic summary: residual distributions and prediction correlations across the nine original classical configurations. The correlation matrix describes similarity among prediction vectors; it is not a model-performance metric.

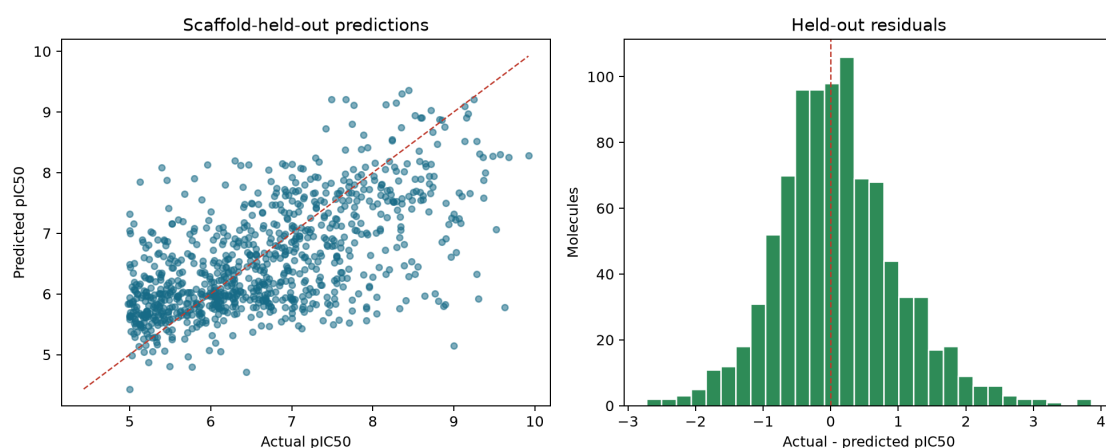


Figure 4: Saved GNN held-out diagnostic output showing predicted versus actual pIC_{50} and the residual distribution for the scaffold-held-out test set.

D Reproducibility Record

Item	Recorded value
GitHub repository	https://github.com/TheHashiramaSenju/AcHe-QSAR-modelling
Active report code-link ref	senju; full SHA recorded below
Saved audit branch / commit	Correction-Branch; full SHA recorded below
Raw data SHA-256	492f42efacc8b49bfa607c0e6ffa66f4f40f5822094f68a54509e58e61ee4418
Final molecular-data SHA-256	b6ad18c926faf5f0221f33342bdc2a105a4bc84471d46ff7c9ed4f496ca20a62
Target	CHEMBL220
Raw endpoint	IC ₅₀
Model target	pIC ₅₀
Final cohort	4,571 molecular identities
Morgan representation	radius 2; 1,024 bits
Descriptor representation	14 descriptors
Original classical split	3,656 train / 915 test
Clean comparison split	3,108 train / 548 validation / 915 test
Original classical CV	5-fold GroupKFold by scaffold
Clean comparison CV	3-fold GroupKFold by scaffold
Original saved Optuna runs	5 completed trials per principal configuration
Clean comparison tuning	8 trials per model/representation
Seed	50
GNN device	CPU
Recorded software	Python 3.12.3; RDKit 2026.03.4; Pandas 2.3.3; NumPy 2.4.3; scikit-learn 1.9.0; SciPy 1.17.1

The exact identifiers are recorded below.

Current GitHub report-link commit: fa675adbeeb801d8291bba5c84f29810ece6d5e3e.

Saved audit commit: bc86f0411a8cd99b408a5a88ae25607cdedee0ad.

Raw-data SHA-256: 492f42efacc8b49bfa607c0e6ffa66f4f40f5822094f68a54509e58e61ee4418.

Final molecular-data SHA-256: b6ad18c926faf5f0221f33342bdc2a105a4bc84471d46ff7c9ed4f496ca20a62.

The report intentionally distinguishes the current GitHub branch used for code hyperlinks from the earlier saved audit snapshot used to preserve data-provenance hashes. This prevents two different repository states from being silently presented as one commit.

D.1 Direct Code Traceability

The small bracketed code references throughout the methods are clickable. For example, [C1] opens the structure-cleaning implementation, [C4] opens the Morgan fingerprint implementation, [C6] opens the original grouped-CV objective, and [C7–C8] open the graph architecture and training implementation. This makes the report auditable at the level of individual methodological

decisions without embedding long source listings in the PDF.

E Complete Model Diagnostics

The following pages reproduce the model-specific diagnostic figures shipped with the project archive. They are included so that the numerical benchmark is accompanied by the actual prediction/error behaviour of every saved classical configuration.

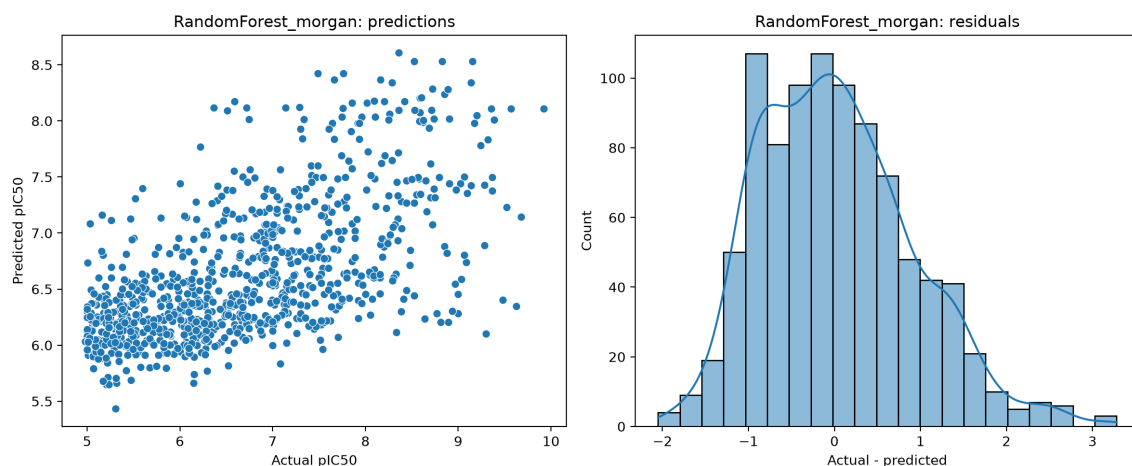


Figure 5: Random Forest with Morgan fingerprints: saved held-out prediction and residual diagnostics.

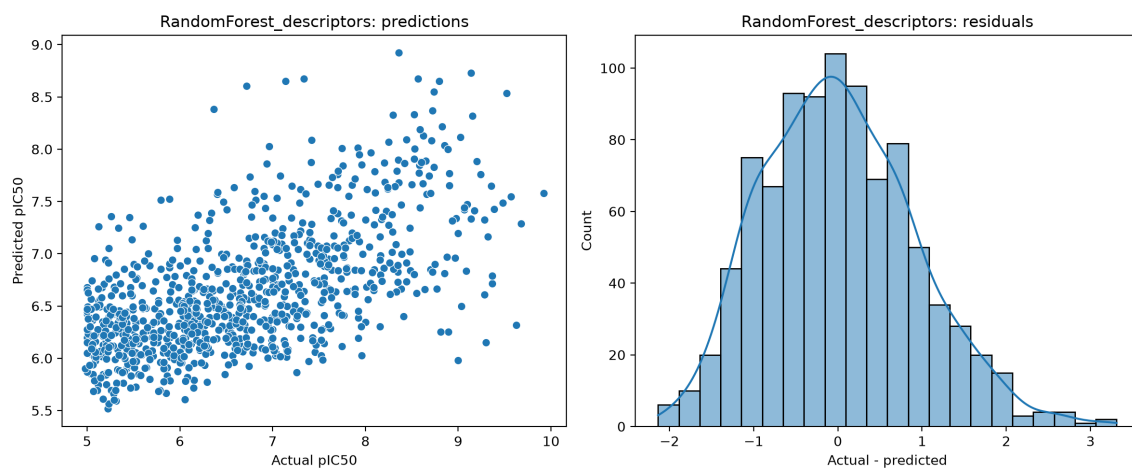


Figure 6: Random Forest with RDKit descriptors: saved held-out prediction and residual diagnostics.

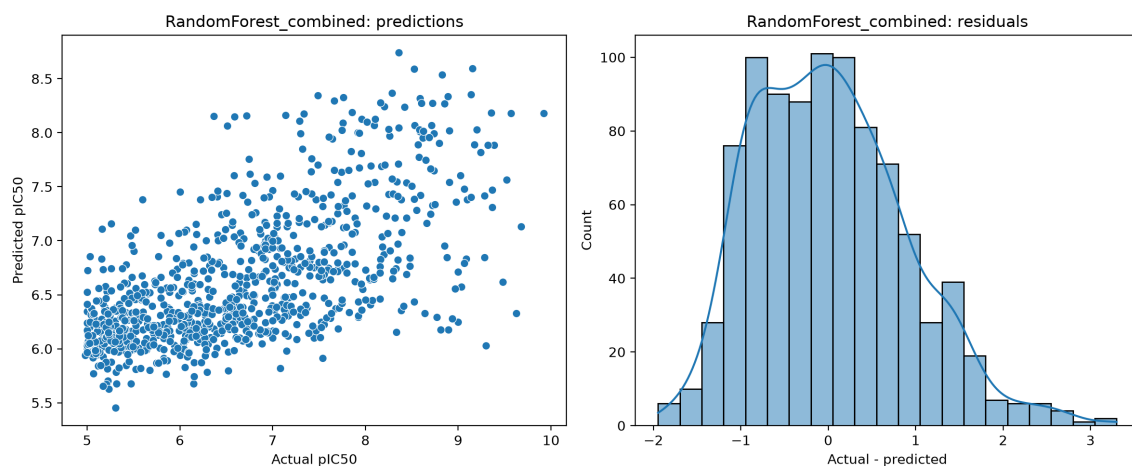


Figure 7: Random Forest with combined Morgan and descriptor features: saved held-out prediction and residual diagnostics.

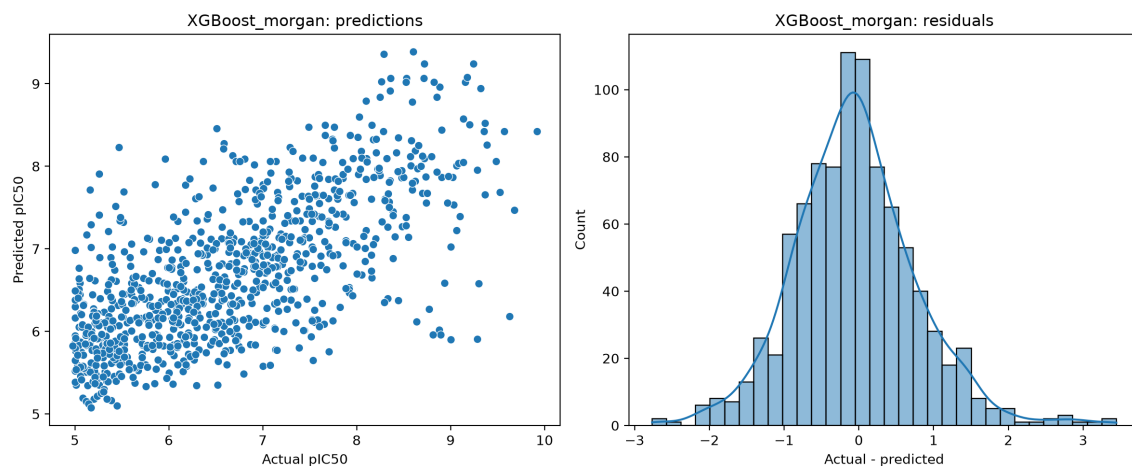


Figure 8: XGBoost with Morgan fingerprints: saved held-out prediction and residual diagnostics.

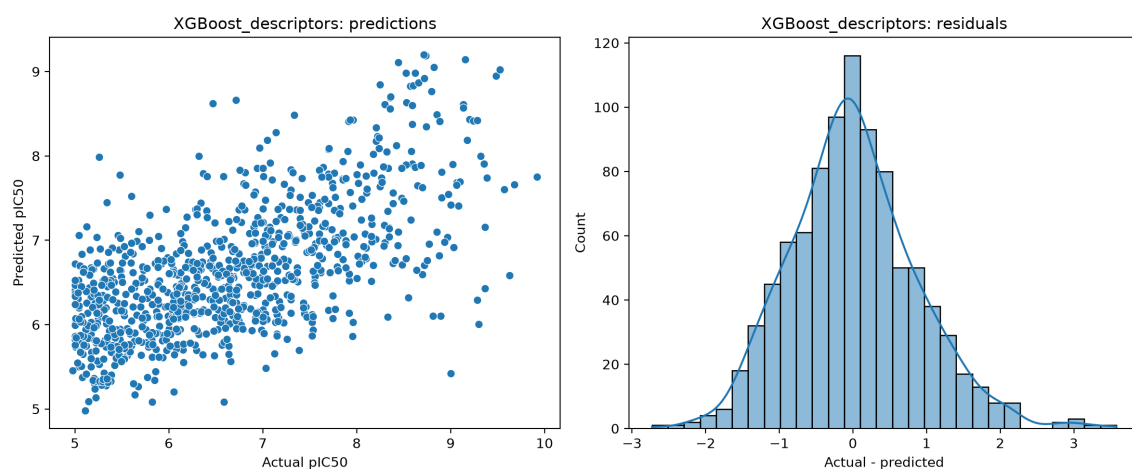


Figure 9: XGBoost with RDKit descriptors: saved held-out prediction and residual diagnostics.

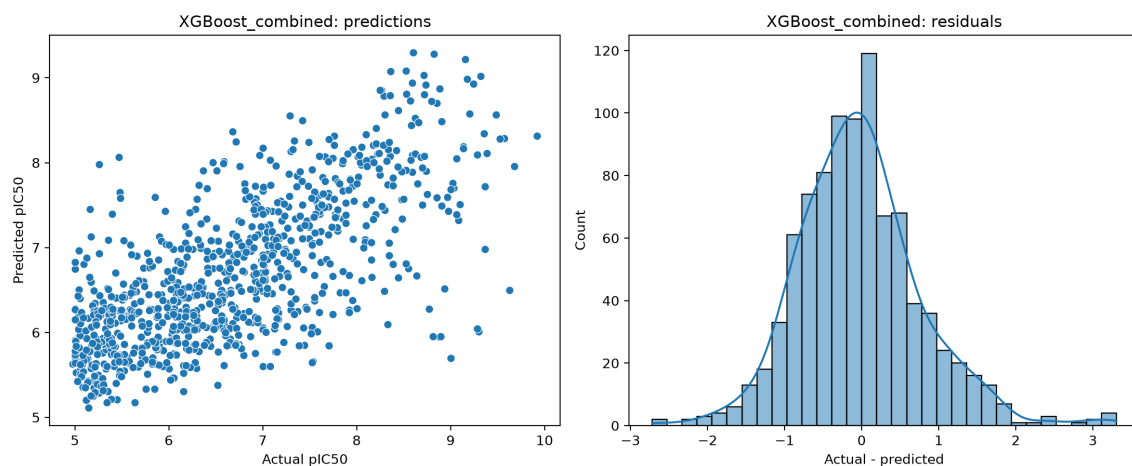


Figure 10: XGBoost with combined features: saved held-out prediction and residual diagnostics; this is the strongest original classical configuration.

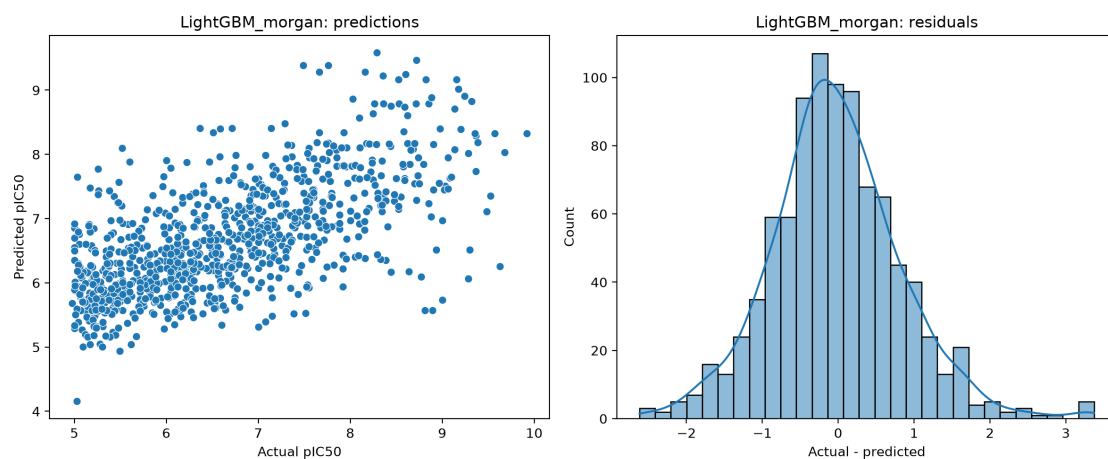


Figure 11: LightGBM with Morgan fingerprints: saved held-out prediction and residual diagnostics.

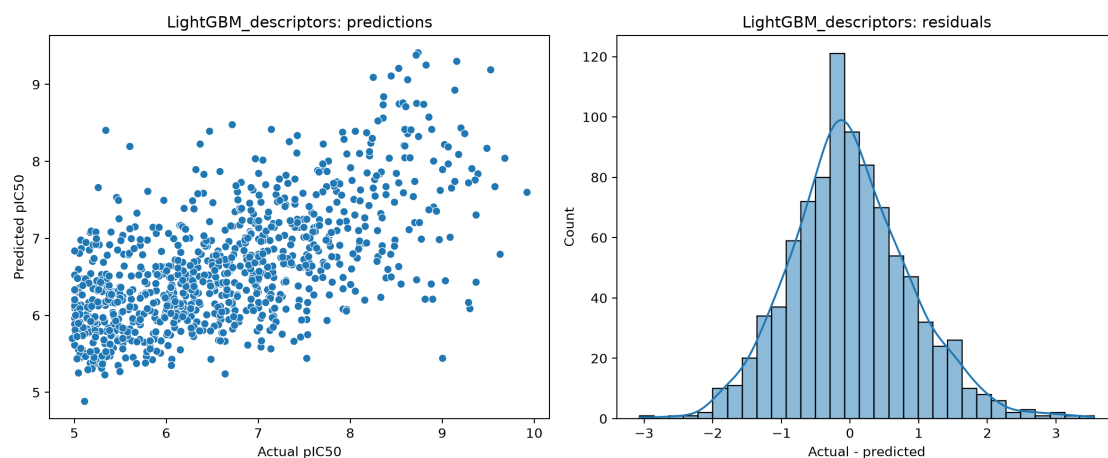


Figure 12: LightGBM with RDKit descriptors: saved held-out prediction and residual diagnostics.

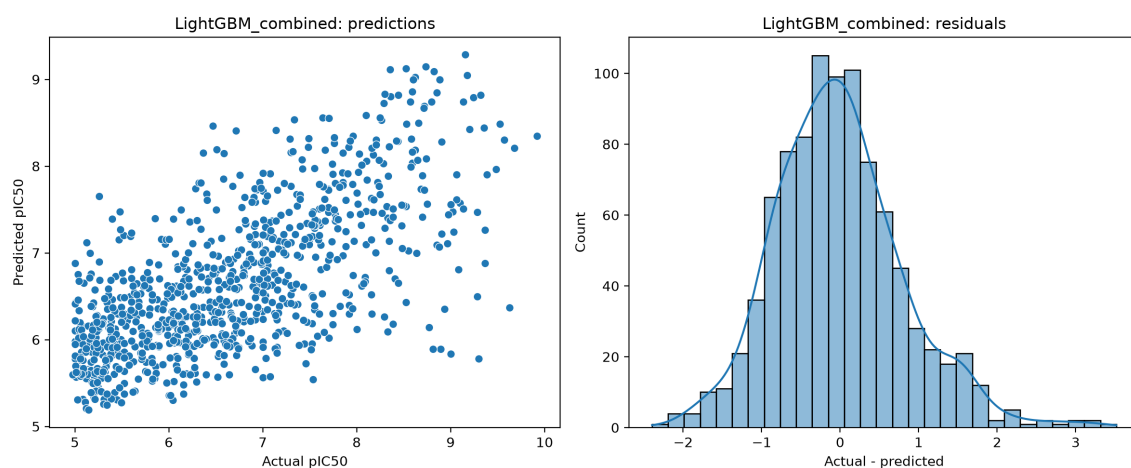


Figure 13: LightGBM with combined features: saved held-out prediction and residual diagnostics.

F Final Audit Record

The final report was checked against the available evidence with the following conclusions:

1. The final curated cohort is 4,571 molecular identities with finite pIC_{50} values.
2. The normalized IC_{50} cutoff is implemented directly in the current cleaning code, and an independent audit found zero rows whose classification differs from a raw-value threshold for this snapshot.
3. The true classical feature dimensions are 1,024 Morgan, 14 descriptors and 1,038 combined. The larger feature counts in original metric CSVs are metadata-reporting errors, not estimator input dimensions.
4. The original scaffold split has zero scaffold and molecule overlap, but the test partition contains singleton scaffolds exclusively; conclusions are therefore limited to this fixed partition.
5. The GNN is a graph-plus-descriptor model with 7 node features, 13 edge channels and 14 global descriptors; it is not graph-only.
6. The saved GNN result is MAE 0.6908, RMSE 0.9093 and $R^2 = 0.3535$ on the 915-molecule held-out test set.
7. The strongest original classical result is XGBoost + combined features with MAE 0.6045, RMSE 0.7981 and $R^2 = 0.5020$.
8. The clean matched audit likewise places combined XGBoost ahead of the GNN on the same 915-molecule test partition.
9. The project contains no completed PiGNN result and none is represented as a completed experiment in this report.
10. No claim of therapeutic efficacy, mechanistic binding prediction, state-of-the-art performance, or universal chemical-space generalization is made.

References

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